

Indian Statistical Institute
Second Semester Examination 2005-2006
B.Math.(Hons.) II Year
Discrete Mathematics & Number Theory

Time: 3 hrs

Date:05-05-06

Max. Marks : 100

Answer all questions.

1. State and prove the quadratic reciprocity law. [10]
2. Find the set of all odd prime numbers p such that 3 is a quadratic residue modulo p . [10]
3. Define a Hadamard matrix. Construct Payley Hadamard matrix. Verify the Hadamard property. [10]
4. Define an extendable 2-design. Show that the design whose points and blocks are, respectively, the points and the hyperplanes of a projective space of dimension d over the finite field $\mathbb{F}_q$ is not extendable if $d > 2$ and $q > 2$. [12]
5. Show that in a projective plane of even order n , any $(n+1)$ -arc extends. Verify that the points of the conic $xz = y^2$ (that is, the set of all roots of this equation) in the projective plane of order 2^t is a $(1 + 2^t)$ -arc. Find an extension of it. [8+8+4]
6. Define an idempotent of a ring. Show that a cyclic code of length n over $\mathbb{F}_q$, $(q, n) = 1$ has a unique, monic, generating idempotent. [8]
7. Define the four binary quadratic residue codes of lengths p , p a prime. Determine their idempotents and their duals. [4+8+8]
8. Find all binary cyclic codes of length 9, their idempotents and their duals. [12]

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